Abelian Varieties

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Lecture 1

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Recall that an *elliptic curve* E over a field k, which is a proper curve which is also a group. In particular, they are projective (in fact, we can write it as a nonsingular plane cubic), and that the group is commutative. If k is algebraically closed, then $E(k)$ is divisible, and for all $n \in \mathbb{N}$,

$$E[n] = \ker(n\cdot : E(k) \to E(k))$$

is finite. In particular, if n is invertible in k, then

$$E[n] = (\mathbb{Z}/n\mathbb{Z})^2$$

On the other hand, if $\mathrm{char}(k) = p > 0$, then either

$$E[p^r] = \mathbb{Z}/p^r\mathbb{Z}$$

for all $r \geq 1$, or

$$E[p^r] = 0$$

for all $r \geq 1$.

The *Abel-Jacobi* theorem gives a homomorphism

$$E(k) \to \mathrm{Cl}^0(E) = \frac{\{\text{degree 0 divisors}\}}{\{\text{principal divisors}\}}$$

Given by $P \mapsto (P) - (0)$. In fact, by Riemann-Roch, this is an isomorphism.

Abelian varieties are "higher dimensional analogues of an elliptic curve". That is, they are a proper variety over k which is a group. If X is an abelian variety, we will see:

- X is projective and nonsingular,
- X is commutative,
- if k is algebraically closed, then $X(k)$ is divisible, with

  $$X[n] \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$$

  where $g = \dim(X)$, n invertible in k. If $\mathrm{char}(k) = p > 0$, then

  $$X[p^r] = (\mathbb{Z}/p^r\mathbb{Z})^t$$

  with $0 \leq t \leq g$.

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- and an analogue of the Abel-Jacobi theorem, there exists another abelian variety $\widehat{X}$, with $\dim(\widehat{X}) = g$, called the dual abelian variety, and an isomorphism

$$\widehat{X}(k) \cong \mathrm{Pic}^0(X) \subseteq \mathrm{Pic}(X)$$

  where $\mathrm{Pic}(X)$ is the group of isomorphism classes of line bundles on X, which is in turn, isomorphic to the divisor class group. Moreover, we have a morphism $X \to \widehat{X}$, which is surjective with finite kernel. Note that we don't need to have $X \cong \widehat{X}$.

In contrast to the one-dimensional case, we need a lot of algebraic geometry, especially the cohomology of coherent sheaves to prove these. One reason is that unlike elliptic curves, we don't have nice equations for them.

Conventions for this course: Every ring will be commutative unital, and ring homomorphisms will be unital. If $f : A \to B$ is a ring homomorphism, we say B is an A-algebra. We'll write ab for $f(a)b$. An A-module is *finite* if it is finitely generated. An A-algebra B is *finite* if it is finite as an A-module, and it is of *finite type* if it is finitely generated as an A-algebra.

A *family* $(x_i)_{i\in I}$ of elements of a set S is just $x_i \in S$, indexed by a set I. This is the same as a function $I \to S$. We'll write $\#S$ for the size of a set S, $\subset$ and $\subseteq$ will be used interchangably. Finally,

$$\mathbb{N} = \{0, 1, \dots\}$$

1 Kähler differentials

Let $\varphi : A \to B$ be a ring homomorphism. We'll define a B-module $\Omega_{B/A}$, which is the module of (Kähler) differentials, with an additive map $\mathrm{d} = \mathrm{d}_{B/A} : B \to \Omega_{B/A}$, such that

$$\begin{aligned} \mathrm{d}(b_1 b_2) &= b_1 \mathrm{d}b_2 + b_2 \mathrm{d}b_1 \\ \mathrm{d}a = \mathrm{d}(\varphi(a)) &= 0 \text{ for all } a \in A \end{aligned}$$

In particular, d is an A-linear map.

Definition 1.1

We define $\Omega_{B/A} = \mathcal{P}/\mathcal{Q}$, where

$$\begin{aligned} \mathcal{P} &= \text{free } B\text{-module on symbols } [b] \text{ for } b \in B \\ \mathcal{Q} &= \text{submodule generated by } [a], [b_1 + b_2] - [b_1] - [b_2], [b_1 b_2] - b_1[b_2] - b_2[b_1] \end{aligned}$$

and

$$\mathrm{d}b = [b] \mod \mathcal{Q}$$

An *A-derivation of B into a B-module M* is an additive map $\mathrm{D} : B \to M$, such that

$$\begin{aligned} \mathrm{D}(A) &= 0 \\ \mathrm{D}(b_1 b_2) &= b_1 D(b_2) + b_2 D(b_1) \end{aligned}$$

We will write $\mathrm{Der}_A(B, M)$ for the set of A-derivations $B \to M$. This is a B-module, with $(b'\mathrm{D})(b) = b'\mathrm{D}(b)$. Moreover, $\mathrm{Der}_A(B, \cdot)$ is a functor. That is, if we have a B-module homomorphism $f : M \to M'$, then we have a map

$$\mathrm{Der}(A, M) \to \mathrm{Der}(A, M')$$

given by $\mathrm{D} \mapsto f \circ \mathrm{D}$. In particular, $(\Omega_{B/A}, \mathrm{d}_{B/A})$ is the *universal* derivation.

Proposition 1.2. Suppose M is a B-module, then there is an isomorphism of B-modules,

$$\begin{aligned} \mathrm{Hom}_B(\Omega_{B/A}, M) &\cong \mathrm{Der}_A(B, M) \\ \psi &\mapsto \psi \circ \mathrm{d}_{B/A} \end{aligned}$$

Proof. The fact that it is a B-module homomorphism is trivial. Let $\mathrm{D} \in \mathrm{Der}_A(B, M)$. Set $\widetilde{\psi} : \mathcal{P} \to M$ to be B-linear, with $\widetilde{\psi}([b]) = D(b)$. Since D is a derivation, $\widetilde{\psi}(Q) = 0$, so we have a map $\psi : \mathcal{P}/\mathcal{Q} = \Omega_{B/A} \to M$, with $\psi \circ \mathrm{d} = D$.

On the other hand, if $\psi \circ \mathrm{d} = 0$, then $\psi = 0$ since $\Omega_{B/A}$ is generated by the image of d. □

Lecture 2

Another characterisation of $\Omega_{B/A}$ is as follows. Consider the map $\mu : B \otimes_A B \to B$, given by $\mu(b_1 \otimes b_2) = b_1 b_2$. In particular, μ is a B-algebra homomorphism, for each of the B-algebra structues on $B \otimes_A B$. Let $J = \ker(\mu)$. Then J/J^2 is a $B \otimes_A B$-module, killed by J, so it is a B-module. In particular, all of the B-module structures on it agree.

Let

$$\begin{aligned} \mathrm{d}' : B &\to J/J^2 \\ b &\mapsto (1 \otimes b - b \otimes 1) \mod J^2 \end{aligned}$$

Then d′ is an A-derivation. To see this,

$$\mathrm{d}'(a) = 1 \otimes a - a \otimes 1 = 0$$

for all $a \in A$. Next,

$$\begin{aligned} b_1 \mathrm{d}'(b_2) &= (b_1 \otimes 1)(1 \otimes b_2 - b_2 \otimes 1) \mod J^2 \\ &= b_1 \otimes b_2 - b_1 b_2 \otimes 1 \mod J^2 \end{aligned}$$

and by symmetry,

$$b_2 \mathrm{d}'(b_1) = 1 \otimes b_1 b_2 - b_1 \otimes b_2 \mod J^2$$

Adding these together gives the Leibniz rule. By the universal property, there exists a unique B-linear map $\psi : \Omega_{B/A} \to J/J^2$, such that $\psi \circ \mathrm{d} = \mathrm{d}'$.

Proposition 1.3. ψ is an isomorphism.

So we could have defined $(\Omega_{B/A}, \mathrm{d})$ as $(J/J^2, \mathrm{d}')$ instead.

Proof. Consider the map

$$\begin{aligned} \varphi : B \otimes_A B &\to \Omega_{B/A} \\ \varphi(b_1 \otimes b_2) &= b_1 \, \mathrm{d}(b_2) \end{aligned}$$

Now

$$J = \left\{ \sum b_i \otimes b_i' \;\middle|\; \sum b_i b_i' = 0 \right\}$$

We claim that J is generated (as a B-module) by elements of the form $1 \otimes b - b \otimes 1$. To see this,

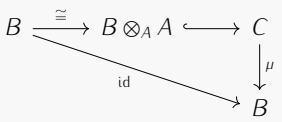
$$\sum b_i \otimes b_i' = \sum (b_i \otimes 1)(1 \otimes b_i' - b_i' \otimes 1) + \sum b_i b_i' \otimes 1$$

and the last term is zero. So it is a B-module under the $b \otimes 1$ action. Now

$$\varphi((1 \otimes b - b \otimes 1)(1 \otimes b' - b' \otimes 1)) = 0$$

and so φ vanishes on J^2, and it factors through $(B \otimes_A B)/J^2$. It remains to check that $\varphi \circ \psi = \mathrm{id}$ and $\psi \circ \varphi|_{J/J^2} = \mathrm{id}$. We leave this as an exercise. □

Remark 1.4. Let $C = (B \otimes_A B)/J^2$, then we have



This gives us an isomorphism of $B = B \otimes_A A$-algebras,

$$B \oplus J/J^2 \cong C$$

The ring structure on the left hand side is given by

$$(b, f) \cdot (b', f') = (bb', bf' + b'f)$$

From the definition, any commutative square of rings

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A' \end{array}$$

induces a map

$$\Omega_{B|A} \to \Omega_{B'|A'}$$

which is B-linear, commuting with d. Moreover, this is transitive for $B \to B' \to B''$.

Proposition 1.5. (i) If $B' = B \otimes_A A'$, then

$$\Omega_{B|A} \otimes_A A' \cong \Omega_{B|A} \otimes_B B' \cong \Omega_{B'|A'}$$

(ii) if $A = A'$, and $B' = S^{-1}B$ is a localisation of B, then

$$\Omega_{S^{-1}B|A} = S^{-1}\Omega_{B|A}$$

Proof. Exercise. □

Example 1.6

If $B = A[t_1, \ldots, t_n]$ is the polynomial algebra, then

$$\Omega_{B|A} = \oplus_{i=1}^{n} B\mathrm{d}t_i$$

is free. To see this,

$$B \otimes_A B = A[\{t_i \otimes 1, 1 \otimes t_j\}] = A[\{t_i \otimes 1, z_i\}] = B[\{z_i\}]$$

where $z_i = 1 \otimes t_i - t_i \otimes 1$. Using this, the map $\mu : B \otimes_A B \to B$ is just $z_i \mapsto 0$ and the identity map on B. So we have that

$$J = \langle z_1, \ldots, z_n \rangle$$

and

$$J^2 = \langle z_i z_j \rangle$$

Thus, J/J^2 is the free module

$$\bigoplus_i B \cdot (z_i \mod J^2)$$

But $z_i \mod J^2$ is just $\mathrm{d}'(z_i) = \mathrm{d}t_i$.

We have two exact sequences for Ω:

Proposition 1.7 (transitivity). if $A \to B \to C$ are ring homomorphisms, then

$$\Omega_{B|A} \otimes_B C \longrightarrow \Omega_{C|A} \longrightarrow \Omega_{C|B} \longrightarrow 0$$

is exact.

Proposition 1.8 (second exact sequence). If $A \to B \twoheadrightarrow C$ are ring homomorphisms, with say $C = B/I$, then

$$I/I^2 \longrightarrow \Omega_{B|A} \otimes C \longrightarrow \Omega_{C|A} \longrightarrow 0$$

is exact, where $f \mod I^2 \mapsto \mathrm{d}f \otimes 1$. This is an exact sequence of C-modules.

Proof of both have been left as exercises.

Corollary 1.9. If $B = A[\{x_i\}]$, and $C = B/\langle\{f_j\}\rangle$, then

$$\Omega_{C/A} = \frac{\bigoplus_i C\,\mathrm{d}x_i}{\sum C\,\mathrm{d}f_j}$$

In this case,

$$\mathrm{d}f_j = \sum_i \frac{\partial f_j}{\partial x_i}\mathrm{d}x_i$$

1.1 Sheafification

Let $f : X \to Y$ be a morphism of schemes. We would like to define a quasicoherent $\mathcal{O}_X$-module $\Omega_{X/Y}$, and a map

$$\mathrm{d} : \mathcal{O}_X \to \Omega_{X/Y}$$

which is $f^{-1}\mathcal{O}_Y$-linear, and satisfies the Leibniz rule for sections. This is called the *sheaf of relative differentials*, or the *cotangent sheaf*. There are two constructions, let's see the 'dirty' one first.

If we have open affines $U = \mathrm{Spec}(B) \subseteq X$, $V = \mathrm{Spec}(A) \subseteq Y$, with $f(U) \subseteq V$, then we can define

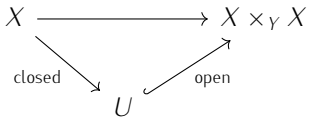
$$\Omega_{X/Y}|_U = \widetilde{\Omega_{B/A}}$$

where $\widetilde{\Omega_{B/A}}$ is the quasicoherent $\mathcal{O}_U$-module attached to the B-module $\Omega_{B/A}$. The map d is induced by

$$\begin{array}{ccc} \mathcal{O}_X(U) & \xrightarrow{\mathrm{d}} & \Omega_{X/Y}(U) \\ \cong\downarrow & & \downarrow\cong \\ B & \xrightarrow[\mathrm{d}]{} & \Omega_{B/A} \end{array}$$

Functoriality means that we can glue to get a quasicoherent sheaf on X. We will omit the details.

For the second construction, consider the diagonal map $\Delta : X \to X \times_Y X$. This map is an immersion, and so it factors as



Let $i : X \hookrightarrow U$ be the closed immersion. Define

$$\Omega_{X/Y} = i^*\left(\mathcal{I}_{X/U}/\mathcal{I}_{X/U}^2\right)$$

where $\mathcal{I}_{X/U}$ is the ideal sheaf. When $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(A)$, this is just $\widetilde{\Omega_{B/A}}$.

Lecture 3

Note that we also have that $\Omega_{X/Y} = i^*\mathcal{I}_{X/U}$, since

$$i^*\mathcal{I}_{X/U} = \mathcal{O}_X \otimes i^{-1}\mathcal{I}_{X/U} = i^{-1}(\mathcal{O}_U/\mathcal{I}_{X/U}) \otimes i^{-1}\mathcal{I}_{X/U}$$

If $X \to Y$ is separated, then we can take $U = X \times_Y X$.

We will define $\mathrm{d} : \mathcal{O}_X \to \Omega_{X/Y}$ by: if we have open affines $V = \mathrm{Spec}(B) \subseteq X$, $U = \mathrm{Spec}(A) \subseteq Y$, with $f(V) \subseteq U$, then

$$\mathrm{d} : \mathcal{O}_X(V) = B \to \Omega_{B/A} = \Omega_{X/Y}(V)$$

is defined. Moreover, this is compatible with restriction to smaller open affines. The set of open affines $\mathrm{Spec}(B) \subseteq X$ such that the image is contained in an open affine in Y, forms a basis for the topology on X. Thus, this extends to a map of sheaves.

Amusing fact: There exists an immersion $X \hookrightarrow Z$ (i.e. a closed immersion then an open immersion), which cannot be factored as an open immersion, then a closed immersion.

We can rewrite the exact sequences, in the setting of sheaves.

- if we have morphisms of schemes $f : X \to Y$, $g : Y \to S$, then we have

$$f^*\Omega_{Y/S} \longrightarrow \Omega_{X/S} \longrightarrow \Omega_{X/Y} \longrightarrow 0$$

- If $Z \hookrightarrow X$ is a closed immersion, with ideal sheaf $\mathcal{I}$, and $f : X \to Y$ a morphism of sheaves, then we have the exact sequence

$$\mathcal{I}/\mathcal{I}^2 \longrightarrow i^*\Omega_{X/Y} \longrightarrow \Omega_{Z/Y} \longrightarrow 0$$

By construction, $\Omega_{X/Y}$ is quasicoherent.

Proposition 1.10. If $f : X \to Y$ is of finite type, and X and Y are Noetherian, then $\Omega_{X/Y}$ is coherent[a].

[a] i.e. affine locally it is the sheaf associated to a finite module.

Proof. Locally, we can cover X with $\mathrm{Spec}(B)$, such that $f(\mathrm{Spec}(B)) \subseteq \mathrm{Spec}(A)$, and this makes B a finite type A-algebra. That is,

$$B = \frac{A[t_1, \dots, t_m]}{I}$$

where $I = \langle f_1, \dots, f_n \rangle$. The second exact sequence is

$$I/I^2 \longrightarrow \Omega_{A[\mathbf{t}]/A} \otimes_{A[\mathbf{t}]} B \longrightarrow \Omega_{B/A} \longrightarrow 0$$

But note that

$$\Omega_{A[\mathbf{t}]/A} \otimes_{A[\mathbf{t}]} B = \bigoplus_i B\mathrm{d}t_i$$

and so $\Omega_{B/A}$ is a finite B-module, and so $\Omega_{X/Y}$ is coherent. □

For field extensions, Ω tells us whether the extension is separable.

Example 1.11

Let L/K be a finite field extension. Then L/K is separable if and only if $\Omega_{L/K} = 0$. To see this, factor the extension as $K \subseteq K_1 \subseteq L$, with K_1/K separable, and L/K_1 purely inseparable. That is, K_1 is the set of all elements of L which is separable over K.

By the primitive element theorem, $K_1 = K(\alpha) = K[t]/\langle g \rangle$, where $g \in K[t]$ is irreducible, $g(\alpha) = 0$ and $g'(\alpha) \neq 0$. So

$$\Omega_{K_1/K} = \frac{K_1\mathrm{d}\alpha}{K_1 g'(\alpha)\mathrm{d}\alpha}$$

by corollary 1.9. But this is zero as $g'(\alpha) \neq 0$. In this case, transitivity gives us $\Omega_{L/K} = \Omega_{L/K_1}$. If $L = K_1$, i.e. L/K is separable, then $\Omega_{L/K_1} = 0$.

Conversely, if L/K is inseparable, then there exists K_2 with $K_1 \subseteq K_2 \subseteq L$, such that $L = K_2(\beta) = K_2[t]/\langle t^p - b \rangle$, with $p = \mathrm{char}(K) > 0$, $b \in K_2$ is not a p-th power. In this case.

$$\Omega_{L/K_2} = L\mathrm{d}\beta$$

since in this case, $\frac{\mathrm{d}}{\mathrm{d}t}(t^p - b) = 0$. Thus, by transitivity again, $\Omega_{L/K} \neq 0$.

1.2 Tangent and cotangent spaces

Let X be a k-scheme, locally of finite type. That is, X has an open cover by affines $\mathrm{Spec}(A)$, where A is a finite type k-algebra. Let $|X|$ denote the set of closed points of X. If $x \in |X|$, then the residue field at k,

$$\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$$

is a finite extension of k. Define

$$\Omega_{X/k}(x) = \Omega_{X/k,x} \otimes_{\mathcal{O}_{X,x}} \kappa(x)$$

This is a finite dimensional $\kappa(x)$ vector space. This follows from the fact that $\Omega_{X/k}$ is coherent. We call this the *cotangent space of X at x*.

The corresponding *tangent space* is

$$T_{X,x} = \mathrm{Hom}_{\kappa(x)}(\Omega_{X/k}(x), \kappa(x))$$

Proposition 1.12. If $\kappa(x) = k$, then there exists a canonical isomorphism

$$\Omega_{X/k}(x) \cong \mathfrak{m}_x/\mathfrak{m}_x^2$$

of k-vector spaces.

Proof. In this case,

$$\Omega_{X/k}(x) = \Omega_{\mathcal{O}_{X,x}/k} \otimes_{\mathcal{O}_{X,x}} k$$

The second exact sequence gives us that

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \Omega_{\mathcal{O}_{X,x}/k} \otimes k \longrightarrow \Omega_{k/k} = 0 \longrightarrow 0$$

It suffices to show that the first map is injective, which in turn, means that it suffices to show that the dual

$$\mathrm{Hom}_{\mathcal{O}_{X,x}/k}(\Omega_{\mathcal{O}_{X,x}/k}, k) \to \mathrm{Hom}_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k)$$

is surjective. The left hand side can be identified with

$$\mathrm{Der}_k(\Omega_{X,x}, k)$$

and so the map is given by

$$D \mapsto \left(f + \mathfrak{m}_x^2 \mapsto D(f) \mod \mathfrak{m}\right)$$

Since $D(\mathfrak{m}_x^2) = 0$ by Leibniz. As $\mathcal{O}_{X,x}/\mathfrak{m}_x = k \hookrightarrow \mathcal{O}_{X,x}$, we must have that

$$\mathcal{O}_{X,x}/\mathfrak{m}^2 = k \oplus \mathfrak{m}_x/\mathfrak{m}_x^2$$

as k-vector spaces. Let $\pi : \mathcal{O}_{X,x}/\mathfrak{m}_x^2 \to \mathfrak{m}_x/\mathfrak{m}_x^2$ be the projection. For $\varphi : \mathfrak{m}_x/\mathfrak{m}_x^2 \to k$, the map $D = \varphi \circ \pi$ is a derivation which maps to φ.

To show that it is a derivation, recall the ring structure on $k \oplus \mathfrak{m}_x/\mathfrak{m}_x^2$ from before. In particular,

$$(a, b)(a', b') = (aa', ab' + a'b)$$

□

Lecture 4

Remark 1.13. THe statement holds more generally, if $\kappa(x)$ is separable over k.

This leads to a "geometric" interpretation of the tangent space.

Proposition 1.14. Let $x \in |X|$, X locally of finite type over k, and suppose $\kappa(x) = k$. Then

$$T_{X,x} \cong \left\{\text{morphisms of } k\text{-schemes } \mathrm{Spec}\left(\frac{k[\varepsilon]}{\langle \varepsilon^2 \rangle}\right) \to X \text{ whose image is } X\right\}$$

The ring $k[\varepsilon]/\langle \varepsilon^2 \rangle$ is called the ring of *dual numbers*, and $\mathrm{Spec}(k[\varepsilon]/\langle \varepsilon^2 \rangle) = \{\langle \varepsilon \rangle\}$. The map $\mathrm{Spec}(k) \to \mathrm{Spec}(k[\varepsilon]/\langle \varepsilon^2 \rangle)$ is given by sending ε to 0. This is a closed immersion.

Morally, $\mathrm{Spec}(k[\varepsilon]/\langle \varepsilon^2 \rangle)$ is "a point with a direction".

Proof. Giving such a morphism is equivalent to giving a local homomorphism

$$\varphi : \mathcal{O}_{X,x} \to \frac{k[\varepsilon]}{\langle \varepsilon^2 \rangle}$$

of k-algebras. But $\mathcal{O}_{X,x} = k \oplus \mathfrak{m}_x$, since k is the residue field. So φ is determined by its restriction to $\mathfrak{m}$, which is a k-linear map $\mathfrak{m}_x \to k\varepsilon \subseteq k[\varepsilon]/\langle \varepsilon^2 \rangle$ (as it is a *local map*).

Since $\varepsilon^2 = 0$, φ is zero on $\mathfrak{m}_x^2$. So it is a k-linear map

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \to k\varepsilon \cong k$$

So the set of such morphisms is isomorphic to

$$\mathrm{Hom}_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k) \cong \mathrm{Hom}_k(\Omega_{X/k}(x), k) = \mathrm{T}_{X,x}$$

□

As above, this is also true when $\kappa(x)/k$ is separable.

Aside: In the proofs of the above, we used the fact that if $\kappa(x) = k$, then

$$\mathcal{O}_{X,x} = k \oplus \mathfrak{m}_x$$

If $\kappa(x) \neq k$, in general, $\mathcal{O}_{X,x}$ will not be a $\kappa(x)$-algebra. But it is the case (at least) when $\kappa(x)/k$ is separable. But

$$\frac{\mathcal{O}_{X,x}}{\mathfrak{m}_x^2} \cong \kappa(x) \oplus \frac{\mathfrak{m}_x}{\mathfrak{m}_x^2}$$

That is, there is a field inside $\mathcal{O}_{X,x}/\mathfrak{m}_x^2$, which maps isomorphically onto $\kappa(x)$. The proof is just Hensel's lemma, write $\kappa(x) = k(\alpha)$, with $g(\alpha) = 0$. Lift $g(\alpha)$ to a solution of $g(t) \equiv 0 \mod \mathfrak{m}^2$.

As an example, let $X = \mathbb{A}_1^{\mathbb{Q}} = \mathrm{Spec}(\mathbb{Q}[t])$, and let $x = \langle t^2 + 1 \rangle \in X$. Then $\kappa(x) = \mathbb{Q}(i)$. But $\mathbb{Q}(i)$ can't be a subfield of the local ring

$$\mathcal{O}_{X,x} \subseteq \mathbb{Q}(t)$$

But the completion

$$\varprojlim_n \frac{\mathcal{O}_{X,x}}{\mathfrak{m}_x^n}$$

does contain a copy of $\kappa(x)$.

1.3 Nonsingular varieties

Let X be an integral scheme of finite type over a field k. Let $d = \dim(X)$.

We say that X is *smooth* over k if $\Omega_{X/k}$ is locally free of rank d. If k is algebraically closed, then this is equivalent to the other conditions:

Theorem 1.15. Suppose X is an integral scheme of dimension d, and of finite type over an algebraically closed field k. Then the following are equivalent.

(i) X is smooth over k,

(ii) for all $x \in |X|$, $\dim_k(\mathrm{T}_{X,x}) = d$,

(iii) for all $x \in |X|$, the local ring $\mathcal{O}_{X,x}$ is *regular*[a] (of dimension d).

[a] A Noetherian local domain R of dimension d is regular if it's maximal ideal $\mathfrak{m}$ can be generated by d elements.

Proof. We know $\dim_k(\mathrm{T}_{X,x}) = \dim_k(\mathfrak{m}_x/\mathfrak{m}_x^2)$.

To show (ii) and (iii) are equivalent, note that

$$\mathfrak{m}_x/\mathfrak{m}_x^2 = \mathfrak{m}_x \otimes_{\mathcal{O}_{X,x}} k$$

and $\mathfrak{m}_x$ is a finite $\mathcal{O}_X$-module. Nakayama's lemma implies that if $t_1, \ldots, t_n \in \mathfrak{m}_x$, then $\mathfrak{m}_x = \langle t_1, \ldots, t_n \rangle$ if and only if $t_i + \mathfrak{m}_x^2$ generate $\mathfrak{m}_x/\mathfrak{m}_x^2$ as a k-vector space.

To show (i) implies (ii),

$$T_{X,x} = \mathrm{Hom}_k(\Omega_{X/k}(x), k)$$

this has dimension d as $\Omega_{X/k}$ is locally free of rank d. For (ii) implies (i), let K be the function field[1] of X. Now as k is algebraically closed, K is a finite separable extension of $K_0 = k(t_1, \ldots, t_d)$, for $t_i \in K$ algebraically independent over k. Hence

$$\Omega_{K/k} = \Omega_{K_0/K} \otimes_{K_0} K$$

and so

$$\Omega_{K_0/k} = \bigoplus_i K_0 \mathrm{d}t_i$$

Thus, $\dim_K(\Omega_{K/k}) = d$. But

$$\Omega_{K/k} = \Omega_{X/k,\eta}$$

Lemma 1.16. Let X be as above.

(i) $\mathcal{F}$ a coherent sheaf $\mathcal{O}_X$-module, such that for all closed points $x \in |X|$, $\dim_k(\mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} k)$ is the same as $\dim_k(\mathcal{F}_\eta) = n$, then $\mathcal{F}$ is locally free.

(ii) In general, there exists a non-empty open subscheme $U \subseteq X$, such that for all $x \in |X|$, $\mathcal{F}_x$ is free if and only if $x \in U$.

Remark 1.17. There are two notions of locally free for any A-module M. The first is that $M_\mathfrak{p}$ is free for all $\mathfrak{p} \in \mathrm{Spec}(A)$. The second is that $\widetilde{M}$ is a locally free sheaf on $\mathrm{Spec}(A)$.

These are not equivalent in general. For example, $A = \mathbb{Z}$ and

$$M = \left\{ \frac{m}{n} \;\middle|\; n \text{ squarefree} \right\} \subseteq \mathbb{Q}$$

$M_\mathfrak{p}$ is free, but $\widetilde{M}$ is not locally free.

In the course, we will always use the second one. If A is Noetherian, and M is finite, then the two are equivalent.

Lecture 5

Proof of lemma. For both of these statements, we can assume that $X = \mathrm{Spec}(A)$ is affine, and $\mathcal{F} = \widetilde{M}$, for a finite A-module M. $K = \mathrm{Frac}(A)$, so $n = \dim_K(M \otimes_A K)$.

Let $\mathfrak{m} \in \mathrm{maxSpec}(A)$ be a maximal ideal. By assumption,

$$n = \dim_K(M \otimes (A/\mathfrak{m}))$$

Choose $t_1, \ldots, t_n \in M$, whose images in $M/\mathfrak{m}M$ form a basis. Now consider the natural map[2] $A^n \to M$ from this basis. By Nakayama, the localisation

$$A^n_\mathfrak{m} \to M_\mathfrak{m}$$

is surjective. So the map $K^n \to M \otimes_A K$ is surjective, so it is an isomorphism. Hence the above map is an isomorphism.

We need to show that there exists a nonempty open $U \subseteq \mathrm{Spec}(A)$, such that the map

$$\mathcal{O}_U^n \to \widetilde{M}|_U$$

is an isomorphism. We know that the map $A^n \to M$ is injective, and has finite torsion cokernel N say, since it is an isomorphism after tensoring with K. Then the map

$$\mathcal{O}_U^n \to \widetilde{M}|_U$$

is an isomorphism for $U = X \setminus \mathbb{V}(\mathrm{Ann}(N))$. Now note that $\mathrm{Ann}(N)$ is non-zero, so U is non-empty. Moreover, $\mathrm{Ann}(M) \not\subseteq \mathfrak{m}$, so $\mathfrak{m} \in U$.

For (ii), choose $t_i \in M$ such that the map $K^n \to M \otimes_A K$ is an isomorphism. The corresponding map $A^n \to M$ has finite torsion cokernel as above, and so at a maximal ideal $\mathfrak{m}$, we have an isomorphism $A^n_\mathfrak{m} \to M_\mathfrak{m}$ if and only if $\mathfrak{m} \notin \mathbb{V}(\mathrm{Ann}(\mathrm{coker}(\cdot)))$. □

[1] i.e. the local ring at the unique generic point.

[2] i.e. $(a_1, \ldots, a_n) \mapsto a_1 t_1 + \cdots + a_n t_n$.

□

Remark 1.18. The theorem holds more generally, for k perfect. But it fails for general k. Let $X = \mathrm{Spec}(K)$, where $K = k(t^{1/p})$, $k = \mathbb{F}_p(t)$. Here,

$$\Omega_{K/k} = K(t)\,\mathrm{d}\left(t^{1/p}\right)$$

is non-zero, but K is clearly regular (of dimension 0).

Thus, over non-perfect fields, regular and smoothness are not equivalent.

Theorem 1.19. Let k be algebraically closed, X/k an integral scheme of finite type over k, then there exists a maximal non-empty open $U \subseteq X$ which is smooth over k.

Proof. As in the proof of (ii) implies (i) in the previous theorem, we know that

$$\Omega_{X/k,\eta} \cong \kappa(X)^d$$

The second part of the lemma says that there exists a non-empty open U, such that for all $x \in |X|$, $x \in U$ if and only if $\Omega_{X/k,x}$ is free of rank d. But by the theorem, this is the same as

$$\dim_k(T_{X,x}) = d$$

□

1.4 Digression - some general (nice) properties of schemes

Let X be a scheme. Recall:

- X is *quasicompact* if X is a finite union of open affines.

  Example 1.20 (non-example)
  Let $X = \mathbb{A}^\infty_k = \mathrm{Spec}(k[x_1, x_2, \dots])$, and $U = X \setminus \{0\}$. This is not quasicompact.

- X is *separated* if the diagonal map is closed. In this case, the intersection of open affines is an open affine.

  Example 1.21 (non-example)
  The bug-eyed plane is not separated.

- (for completeness) X is *quasi-separated* if the intersection of any two open affines is quasicompact.

  Example 1.22 (non-example)
  Let X be the union of two copies of $\mathbb{A}^\infty_k$, glued on the complement of the origin.

  Sometimes, people will say "X is *qcqs*", to say X is quasicoherent and quasiseparated.
  If $f : X \to Y$ is a morphism,

  - it is *quasicompact*, if for all open affines $V \subseteq Y$, $f^{-1}(V)$ is quasicompact,
  - it is *locally of finite type*, if for all $x \in X$, there exists an open affine neighbourhoods $x \in U = \mathrm{Spec}(B)$, and $f(U) \subseteq V = \mathrm{Spec}(A)$, where B is a finite type A-algebra.
  - it is of *finite type* if it is quasicompact and locally of finite type.
  - (for completeness) it is *locally of finite presentation* if locally it is $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$, where B is a finitely presented A-algebra. So
    $$B = \frac{A[x_1, \dots, x_n]}{\langle f_1, \dots, f_m \rangle}$$
    When A is Noetherian, this is the same as locally of finite type. That is, when Y is locally Noetherian. In this case, f is *of finite presentation* if moreover it is quasicompact and quasiseparated.

2 Flatness and related notions

Let A be a ring. An A-module M is *flat over* A if for every injection $N \to N'$ of A-modules, the corresponding map $M \otimes_A N \to M \otimes_A N'$ is injective.

We say that an A-algebra B is *flat* if it is flat as an A-module.

Proposition 2.1. Some facts:

(i) A is flat over itself,

(ii) any free module is flat, so vector spaces over a field are flat,

(iii) M is flat if and only if for all ideals $I \trianglelefteq A$, the map $M \otimes_A I \to M$ is injective,

(iv) M is flat if and only if for every short exact sequence

$$0 \longrightarrow N_1 \longrightarrow N_2 \longrightarrow N_3 \longrightarrow 0$$

The corresponding sequence

$$0 \longrightarrow M \otimes_A N_1 \longrightarrow M \otimes_A N_2 \longrightarrow M \otimes_A N_3 \longrightarrow 0$$

is still exact. That is, $M \otimes_A \cdot$ is an *exact functor*. Note that it is always exact except at $M \otimes_A N_1$. So tensoring is a *right exact functor*.

(v) if M is A-flat, $A \to B$ is a ring homomorphism, then $M \otimes_A B$ is B-flat. This follow from the fact that

$$(M \otimes_A B) \otimes_B N = M \otimes_A N$$

(vi) M is A-flat if and only if $M_{\mathfrak{p}}$ is $A_{\mathfrak{p}}$-flat for all $\mathfrak{p} \in \operatorname{Spec}(A)$. This follows from the fact that A-modules are the same as quasicoherent sheaves on $\operatorname{Spec}(A)$, and exactness of sheaves is equivalent to exactness on stalks.

(vii) if A is a PID (or more generally, a Dedekind domain), then M is flat if and only if it is torsion-free. By the above, we can reduce to the case of $M_{\mathfrak{p}}$ being $A_{\mathfrak{p}}$-flat. But $A_{\mathfrak{p}}$ is (a field or) a discrete valuation ring. The result then follows from (iii), since in a DVR, $I = \langle \pi^n \rangle$ for some n.

(viii) Let A be Noetherian, M a finite A-module. Then the following are equivalent.

- M is flat,
- M is projective,
- $M_{\mathfrak{p}}$ is free over $A_{\mathfrak{p}}$ for all $\mathfrak{p} \in \operatorname{Spec}(A)$,
- $\widetilde{M}$ is locally free. That is, there exists $f_1, \ldots, f_n \in A$, such that $\langle f_1, \ldots, f_n \rangle = A$, and M_{f_i} is free over A_{f_i} for all i.

So over a Noetherian ring, finite flat is the same as being finite and locally free.

(ix) Since localisation is exact, it preserves flatness.

Lecture 6

Proposition 2.2. Let

$$\cdots \longrightarrow K^{n-1} \xrightarrow{\mathrm{d}^{n-1}} K^n \xrightarrow{\mathrm{d}^n} K^{n+1} \longrightarrow \cdots$$

be a complex of A-modules. Now suppose we have an A-module M, then we have a natural map

$$\mathrm{H}^n(K^\bullet, d^\bullet) \otimes_A M \to \mathrm{H}^n(K^\bullet \otimes_A M, \mathrm{d}^\bullet \otimes \mathrm{id})$$

which is an isomorphism if M is flat.

Proof. The sequence

$$0 \longrightarrow \ker(\mathrm{d}^n) \hookrightarrow K^n \longrightarrow K^{n+1}$$

is exact, so we get

$$\ker(\mathrm{d}^n) \otimes_A M \to \ker(\mathrm{d}^n \otimes \mathrm{id}_M) \subseteq K^n \otimes_A M \tag{$*$}$$

and we have

$$K^{n-1} \xrightarrow{\mathrm{d}^{n-1}} \ker(\mathrm{d}^n) \longrightarrow \mathrm{H}^n(K^\bullet, d^\bullet) \longrightarrow 0$$

which is also exact. So the rows of

$$\begin{array}{ccccccc} K^{n-1} \otimes_A M & \longrightarrow & \ker(\mathrm{d}^n) \otimes_A M & \longrightarrow & \mathrm{H}^n(K) \otimes_A M & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ K^{n-1} \otimes_A M & \longrightarrow & \ker(\mathrm{d}^n \otimes_A \mathrm{id}_M) & \longrightarrow & \mathrm{H}^n(K \otimes_A M) & \longrightarrow & 0 \end{array} \tag{$**$}$$

are exact. Hence we have an induced map $\mathrm{H}^n(K) \otimes M \to \mathrm{H}^n(K \otimes M)$. If M is flat, then the map $\ker(\mathrm{d}_n) \otimes_A M \to \ker(\mathrm{d}^n \otimes_A \mathrm{id}_M)$ is an isomorphism. Hence by the five lemma, the map $\mathrm{H}^n(K) \otimes_A M \to \mathrm{H}^n(K \otimes_A M)$ is an isomorphism. □

Returning to (algebraic) geometry: Let $f : X \to Y$ be a morphism of schemes. We say that f is *flat* if one of the following (equivalent) conditions are satisfied:

- locally, f is of the form $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$, where B is a flat A-algebra,
- for all $x \in X$, $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$.

More generally, a queasicoherent $\mathcal{O}_X$-module $\mathcal{F}$ is *flat over* Y if for all $x \in X$, $\mathcal{F}_x$ is a flat $\mathcal{O}_{Y,f(x)}$-module.

Example 2.3

Open immersions are flat, since locally, they are isomorphisms. Closed immersions are in general, not flat.

Let X, Y be k-schemes. Then

$$X \times_{\mathrm{Spec}(k)} Y \to Y$$

is flat. Morally, flatness is saying that "the fibres vary continuously".

See more in section in Hartshorne on flat morphisms, especially Example III.9.8.4, of a particular flat family of curves.

Fact: Let $f : X \to Y$ be a flat morphism of integral k-schemes of finite type. Then for all $y \in Y$, and every irreducible component $Z \subseteq f^{-1}(y)$, $\dim(Z)$ is independent of choice of y, Z, and is equal to $\dim(X) - \dim(Y)$.

Example 2.4

Let $Y = \mathbb{P}^2_k$, and let X be the blowup of Y at a point $y \in Y(k)$[a]. Then for all $y' \neq y$, $f^{-1}(y')$ is a point, as we have an isomorphism $X \setminus f^{-1}(y) \cong Y \setminus \{y\}$. But $f^{-1}(y) = \mathbb{P}^1_k$. So f is not flat.

[a] $Y(k) = \mathrm{Hom}_{\mathrm{Spec}(k)}(\mathrm{Spec}(k), Y)$ is the set of k-points of Y.

As an example of blowup for $\mathbb{A}^2$, consider

$$\{(x_0, x_1), (y_0 : y_1) \mid x_0 y_1 = x_1 y_0\} \subseteq \mathbb{A}^2 \times \mathbb{P}^1$$

and the projection onto the $\mathbb{A}^2$-factor. Away from the origin, the fibre is a line. At the origin, the fibre is a $\mathbb{P}^1$.

Lecture 7

One way to think about this is that fibre dimension is constant across a flat family.

Example 2.5

Let $Y = \mathrm{Spec}(k[t])$, and take $X = \mathrm{Spec}(k[x, y, t]/\langle xy - t \rangle)$. We have a ring map $k[t] \to k[x, y, t]/\langle xy - t \rangle$, so we have a scheme map $X \to Y$.

When a is non-zero, the fibre is the (smooth irreducible) conic $xy = a$. When $a = 0$, then we have the singular curve $xy = 0$, and the fibre is reducible.

If instead we considered $X = \mathrm{Spec}(k[x, y, t]/\langle x^2 - ty \rangle)$. When $t = 0$, we have $x^2 = 0$, which is a

non-reduced scheme.

We have the very useful result:

Theorem 2.6 ((a special case of) miracle flatness)**.** Let X, Y be integral k-schemes of finite type and smooth over k. Let $f : X \to Y$ be a morphism. Suppose that for all $y \in Y$, and for all irreducible components $Z \subseteq f^{-1}(y)$, $\dim(Z) = \dim(X) - \dim(Y)$. Then f is flat.

This is actually true under much weaker hypotheses on X. See Eisenbud Commutative Algebra for a proof.

Fact: Let $f : X \to Y$ be flat, and locally of finite presentation (for example, if X, Y are Noetherian, then f just has to be flat of finite type). Then f is an open map.

Note that a finiteness condition is necessary, for example, if $X = \mathrm{Spec}(\mathbb{Q}) \hookrightarrow \mathrm{Spec}(Z) = Y$, then this is flat but not open.

3 Sheaf cohomology

In this section, we will assume (unless stated otherwise), all schemes are Noetherian and separated. In particular, they are quasicompact.

3.1 Homological algebra

Recall a *(cochain) complex of abelian groups (or R-modules, or sheaves)* is a sequence of maps $(A^\bullet, \mathrm{d})$

$$A^0 \xrightarrow{\mathrm{d}} A^1 \xrightarrow{\mathrm{d}} \cdots$$

which may or may not be infinite, such that $\mathrm{d}^2 = 0$. Associated to a (cochain) complex is the *cohomology*, which is

$$\mathrm{H}^p(A^\bullet) = \frac{\ker(\mathrm{d} : A^p \to A^{p+1})}{\mathrm{im}(\mathrm{d} : A^{p-1} \to A^p)}$$

We write

$$\mathrm{H}^*(A^\bullet) = \bigoplus_p \mathrm{H}^p(A^\bullet)$$

for the graded R-module. A *morphism of complexes* $f : (A, \mathrm{d}) \to (B, \mathrm{d})$ is a family of maps $f^p : A^p \to B^p$, such that

$$\mathrm{d}f = f\mathrm{d}$$

This induces homomorphisms

$$\mathrm{H}^p(f) : \mathrm{H}^p(A^\bullet) \to \mathrm{H}^p(B^\bullet)$$

Suppose

$$0 \longrightarrow A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \longrightarrow 0$$

is an exact sequence of complexes, then we have homomorphisms $\partial : \mathrm{H}^p(C^\bullet) \to \mathrm{H}^{p+1}(A^\bullet)$, such thats

$$0 \longrightarrow \mathrm{H}^0(A) \longrightarrow \mathrm{H}^0(B) \longrightarrow \mathrm{H}^0(C) \longrightarrow \mathrm{H}^1(A) \longrightarrow \mathrm{H}^1(B) \longrightarrow \cdots$$

is exact.

Suppose $f, g : A \to B$ are morphisms of chain complexes. They are *homotopic* if there exists maps $h^p : A^p \to B^{p-1}$, such that

$$\mathrm{d}h + h\mathrm{d} = f - g$$

If f and g are homotopic, then they induce the same map on cohomology. In particular, if $A = B$, and $h^p : A^p \to A^{p-1}$ are such that

$$\mathrm{d}h + h\mathrm{d} = \mathrm{id}$$

Then the identity and zero maps on cohomology are equal, so the cohomology is zero. If such an h exists, we say that A is null-homotopic.

Example 3.1

Let A be a cochain complex of abelian groups, $n \geq 0$. Let $\mathbb{Z}[-n]$ be the complex which is $\mathbb{Z}$ in degree n, and zero otherwise. Then the set of morphisms $\mathbb{Z}[-n] \to A$ is the same as the set of maps $f : \mathbb{Z} \to A^n$, with $\mathrm{d}f = 0$. But any such map is of the form $f(k) = kx$ for some $x \in \ker(\mathrm{d} : A^p \to A^{p+1})$. Moreover, the homotopy classes of such maps is the above quotiented by $\mathrm{im}(\mathrm{d} : A^{p-1} \to A^p)$, i.e. $\mathrm{H}^n(A)$. So we obtain the identification

$$\mathrm{H}^n(A) = \{\text{homotopy classes of morphisms } \mathbb{Z}[-n] \to A\}$$

In general, we can have that $\mathrm{H}^*(A) = 0$, with A not null-homotopic. For example, consider

$$\mathbb{Z}/2 \xrightarrow{\cdot 2} \mathbb{Z}/4 \longrightarrow \mathbb{Z}/2 \longrightarrow 0$$

This is exact, but not null-homotopic.

If A is a complex of vector spaces over a field k, then we can always "split off" the cohomology as a direct summand of A^n. In particular, if $\mathrm{H}^n(A) = 0$ for all n, then we can write A as

$$B^0 \longrightarrow B^0 \oplus B^1 \longrightarrow B^1 \oplus B^2 \longrightarrow \cdots$$

The obvious map $h^p : B^{p-1} \oplus B^p \to B^{p-2} \oplus B^{p-1}$ satisfies

$$\mathrm{d}h + h\mathrm{d} = \mathrm{id}$$

and so for complexes of k-vector spaces, $\mathrm{H}^* = 0$ is equivalent to being null-homotopic.

As a special case of the long exact sequence, if we have long exact sequences of length 2, so $A \to A'$ and so on.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0 \end{array}$$

Then the long exact sequence is

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \ker(\alpha) & \longrightarrow & \ker(\beta) & \longrightarrow & \ker(\gamma) & & \\ & & & & & \searrow & & & \\ & & \mathrm{coker}(\alpha) & \longrightarrow & \mathrm{coker}(\beta) & \longrightarrow & \mathrm{coker}(\gamma) & \longrightarrow & 0 \end{array}$$

It is very easy to see that this holds with some of the zeroes removed. This is the *Snake lemma*.